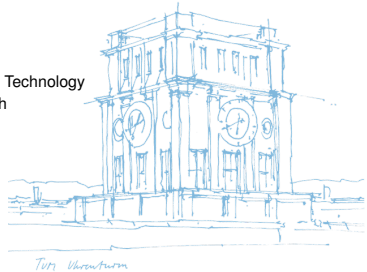


An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

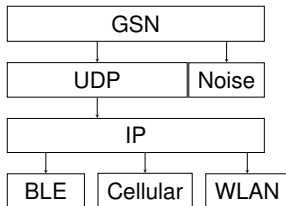
Dan Bachar, David Guzman

Monday 27th April, 2026

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

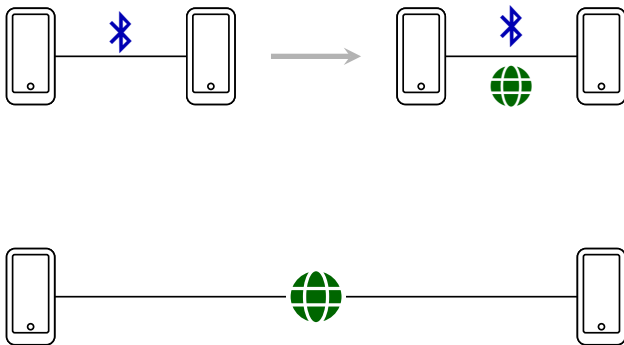


Architecture



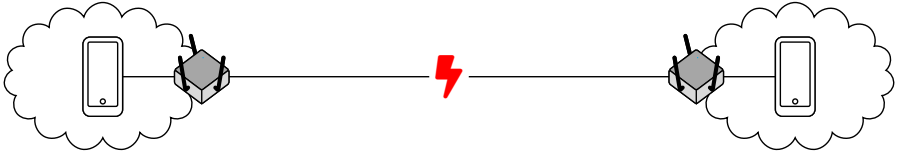
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

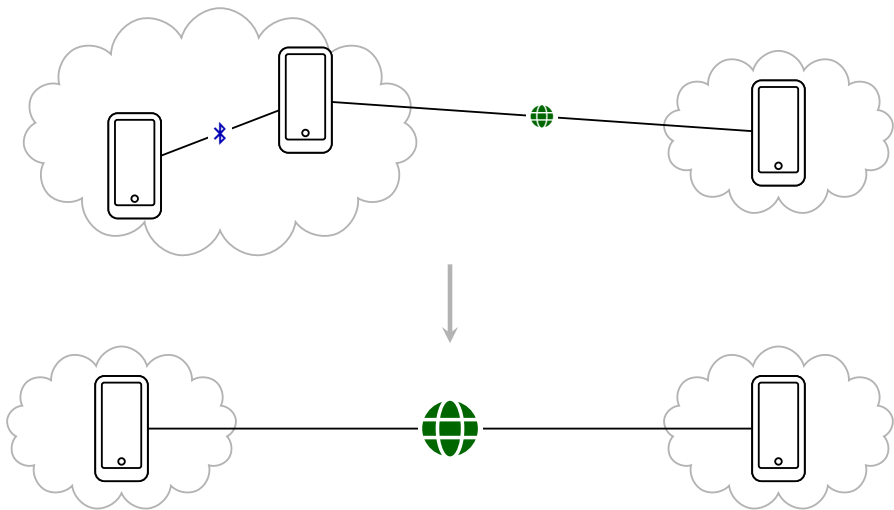


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

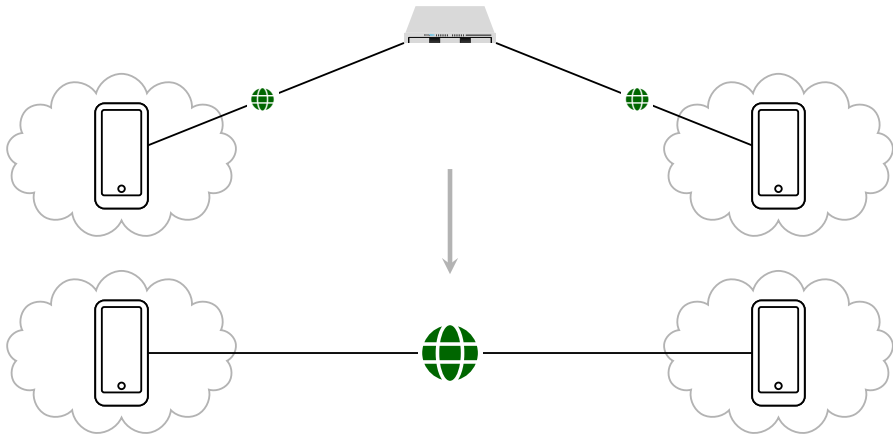
NAT



Well-connected friends



Rendezvous Server



Next steps

- Stabilize connectivity, testing
- Deploy GLP compiled to dart code on smartphones
- Write